Teacher Copy — includes the Answer Key & Teacher Guide. Do not distribute to students.

UNIT 3 · STANDARD 6.AT.1

Understand Ratios

Lesson 3-1 · Teacher Copy

Name: Type your name **Date:** Today's date **Class:** Class period

CHEF ACADEMY MISSION

The Signature Recipe

You just earned your apron at Chef Academy. Head Chef Reyes is trusting you with her signature cookie recipe — 3 cups of flour for every 2 cups of sugar. Before service begins, you must read every recipe like a pro by mastering ratios, or the whole kitchen falls behind.

My Notes

Level 3 Enrichment

🎯 **TODAY'S OBJECTIVES**

Content Objective: I can write and describe a ratio that compares two quantities.

Language Objective: I can explain a comparison using the words ratio, part-to-part, part-to-whole, and colon notation.

📖 See **Learn It** for the explanation and worked examples, and the **Vocab** tab for the words. These are your notes to fill in and keep.

✏️ **Fill in each blank as we go. Use the Word Bank to help you.**

📚 **WORD BANK — FILL EACH BLANK WITH THE BEST WORD**

Ratio | Comparison | Part-to-part | Part-to-whole | Colon notation | Tape diagram | Double number line

👆 *Tap any word to see its meaning and a picture.*

Level 3: try the blanks from memory first, then check the bank.

1. A comparison of two quantities, like 3 to 5, is a _____.

2. Showing how two amounts relate to each other is a _____.

3. A ratio that compares one part of a group to another part is a _____ ratio.

4. A ratio that compares one part of a group to the total is a _____ ratio.

5. Writing a ratio with a colon, like 3:5, is using _____.

6 A bar cut into equal parts that shows how a ratio breaks apart is a _____ _____ .

7 Two lines whose matching ticks pair the two amounts in a ratio form a _____ _____ _____ .

Write About the Math

The school cafeteria is planning a new lunch menu. They want to offer 3 main dishes for every 2 side dishes. If they have 10 side dishes, the cafeteria manager needs to figure out how many main dishes to prepare?

Say your answer to a partner first. Then write it, using at least two of these words:

- ☐ **Ratio** — *Razón*
- ☐ **Comparison** — *Comparación*
- ☐ **Part-to-part** — *Parte a parte*
- ☐ **Part-to-whole** — *Parte a todo*
- ☐ **Colon notation** — *Notación con dos puntos*

Model: *The recipe uses 3 cups of apple juice for every 2 cups of sparkling water.* — A strong answer names that apple juice and sparkling water are being compared and states the relationship as 3 to 2 (or 'for every 3 apple juice, 2 sparkling water'), not just that there is more apple juice.

Start

Complete the frames. Use the word bank.

Completa las oraciones. Usa el banco de palabras.

- **For every ____ main dishes the cafeteria serves ____ side dishes, so with 10 side dishes they need ____ main dishes.**
- **My answer is ____ because ____.**
  Mi respuesta es ____ porque ____.

Build

Write two connected sentences. Use because, so, or but.

Escribe dos oraciones conectadas. Usa porque, entonces o pero.

- **My claim is ____.**
  Mi afirmación es ____.
- **My evidence is ____, so ____.**
  Mi evidencia es ____, entonces ____.

Explain

Write a claim, give math evidence, and explain your reasoning.

Escribe una afirmación, da evidencia matemática y explica tu razonamiento.

- [ ] I answered the question.
- [ ] I used math evidence: numbers, an equation, or a model.
- [ ] I used at least two math words.

Writing Guide — Teacher Copy

- The answer to the math question is correct.
- The evidence is real lesson mathematics (numbers, equation, or model), not restated claim.
- The support level changed the language scaffolding, never the mathematical expectation.

Answer Key & Teacher Guide

1. **Notes 1:** Ratio
2. **Notes 2:** Comparison
3. **Notes 3:** Part-to-part
4. **Notes 4:** Part-to-whole
5. **Notes 5:** Colon notation
6. **Notes 6:** Tape diagram
7. **Notes 7:** Double number line
8. **Watch for this mistake:** *A common mistake in Understand Ratios is writing the ratio in the wrong order: when asked for the ratio of apple juice to sparkling water in a recipe that uses 3 cups of apple juice for every 2 cups of sparkling water, some students write 2:3 (sparkling water to apple juice) because they list the ingredients in whatever order comes to mind rather than the order the question asks for. The two ratios describe different drinks — 3:2 means 3 cups of juice for every 2 cups of water, while 2:3 would mean only 2 cups of juice for every 3 cups of water. Always match the order of your ratio to the order named in the question: "apple juice to sparkling water" means the apple juice number comes first, giving 3:2.*
9. **Try It 1:** A. 5:12 — *The whole class is 5 + 7 = 12 students. Part-to-whole compares one part to the total, so chefs to whole class is 5:12.*
10. **Try It 2:** B. 9:4 — *“For every 9 honey to 4 lemon” is a part-to-part comparison. In colon notation, honey to lemon is 9:4.*